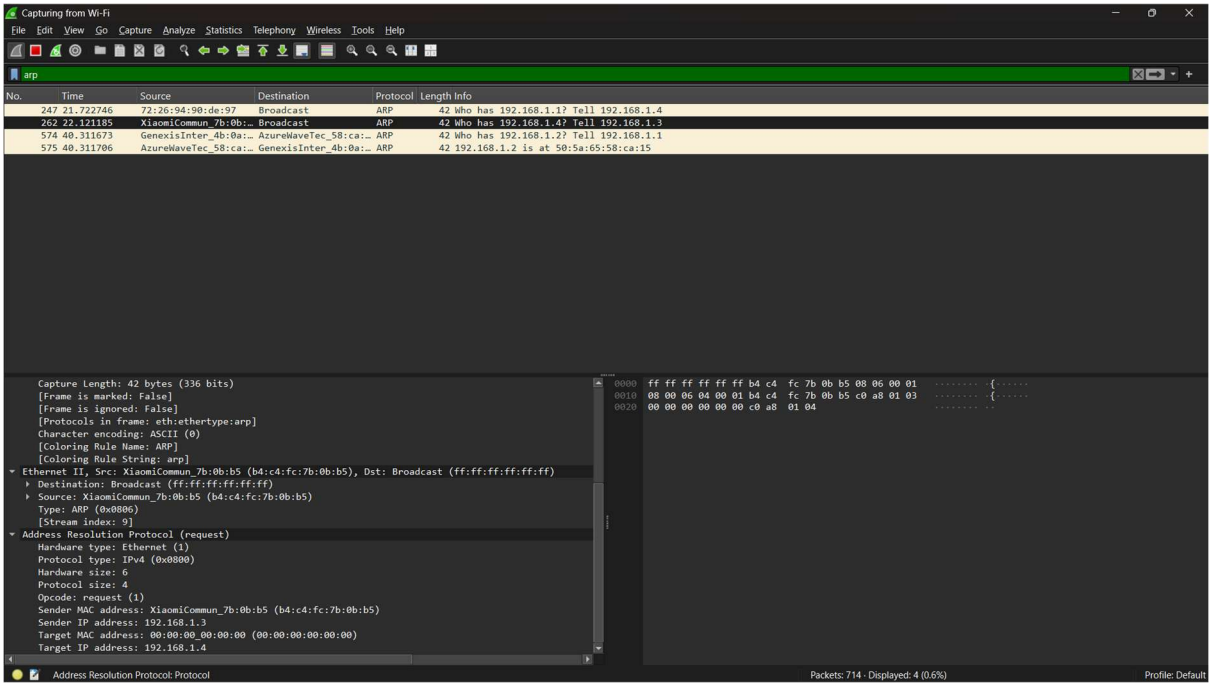
1)



ARP (Address Resolution Protocol) is essential in packet forwarding because it enables the mapping of logical IP addresses to physical MAC addresses required for data transmission within a local network. When a device needs to send a packet, it uses ARP to determine the destination MAC address corresponding to the known IP address, allowing the frame to be correctly delivered at the data link layer. This is particularly important when forwarding packets to a default gateway for communication with external networks. Without ARP, devices would be unable to identify the correct hardware address, and packet delivery would fail even if the IP address is known.

2)

3) 10.0.0.0/24 , 4 Subnets

$2^n = 4$

$\boxed{n = 2}$

$/24 \rightarrow /24+2 = /26$

$\therefore$ Subnet = 255.255.255.192

Total possible subnets = /32

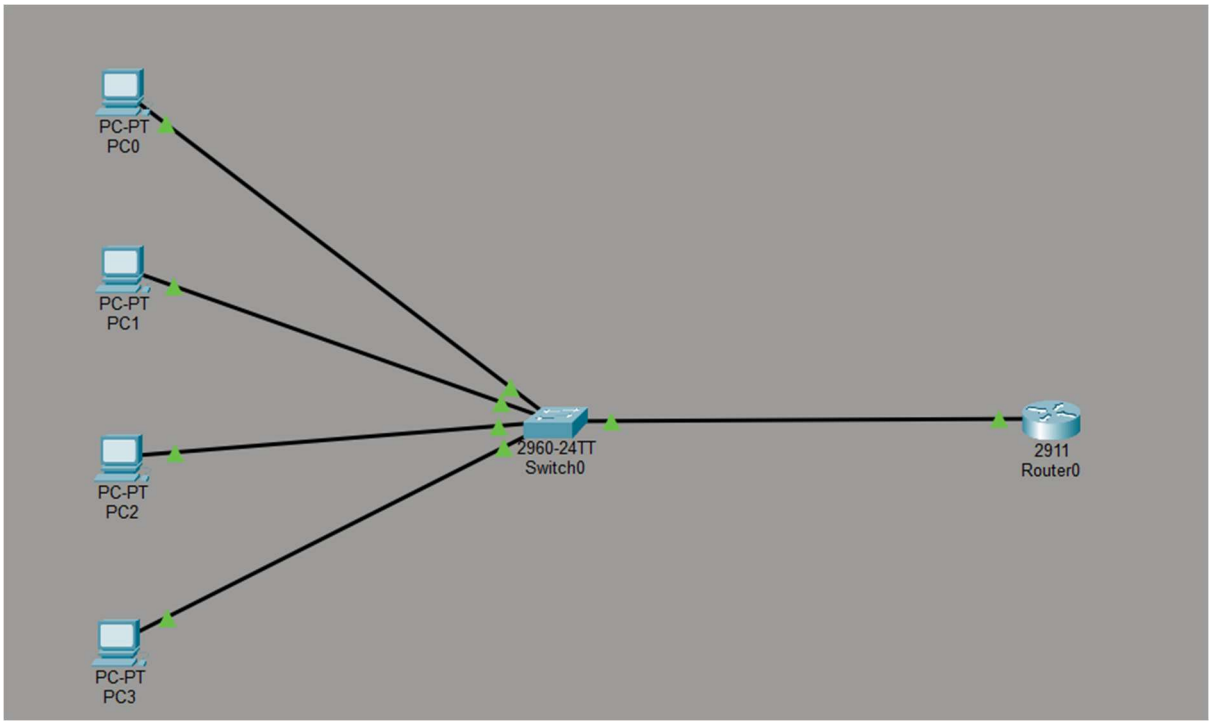
Available $= (2^{32-26}) = 2^6 = 64$

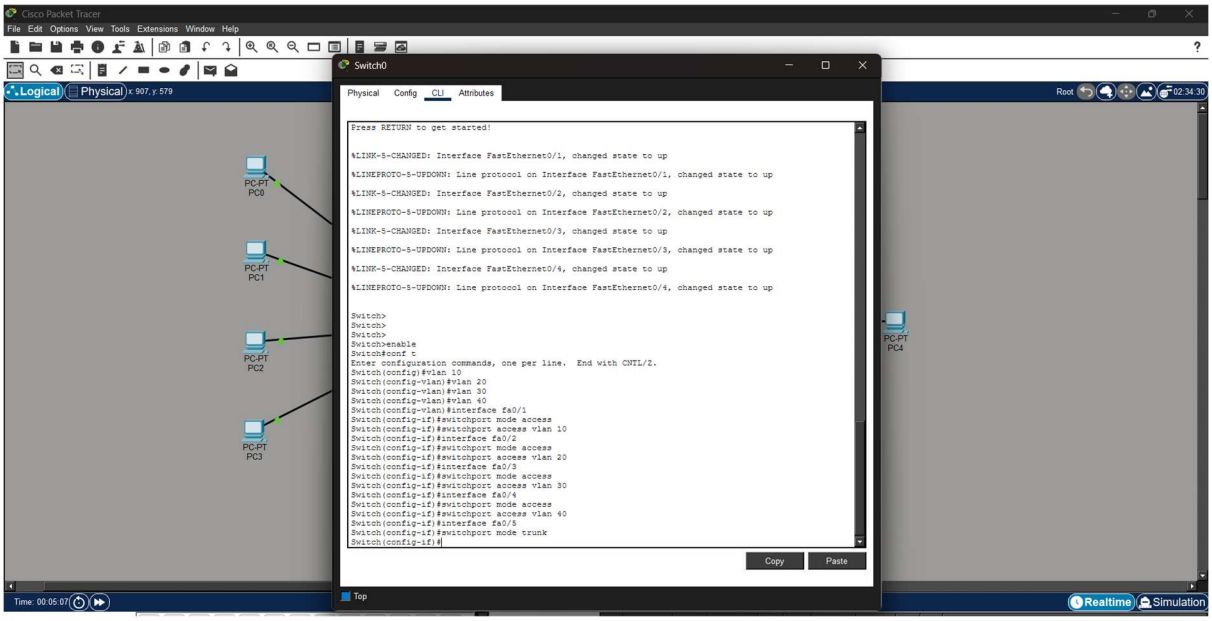
N/W ID	IP - RANGE	Broadcast
10.0.0.0/26	10.0.0.1 - 10.0.0.62	10.0.0.63
10.0.0.64/26	10.0.0.65 - 10.0.0.126	10.0.0.127
10.0.0.128/26	10.0.0.129 - 10.0.0.190	10.0.0.191
10.0.0.192/26	10.0.0.193 - 10.0.0.254	10.0.0.255

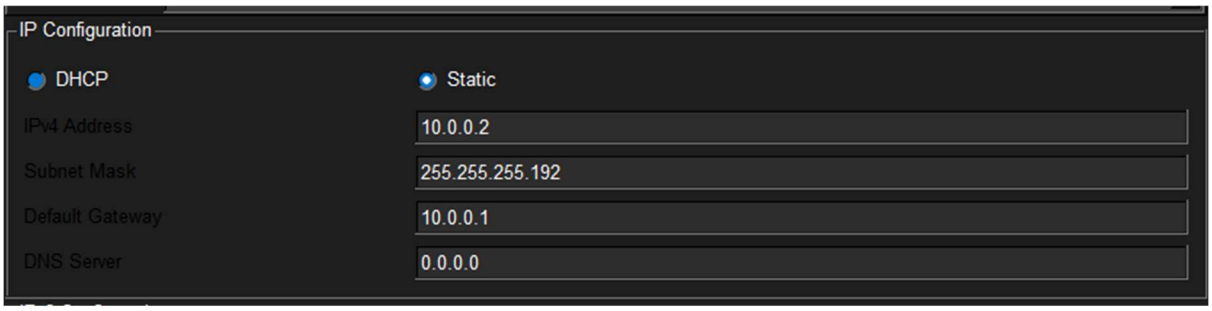
3)

The four subnets are:

1. Network: 10.0.0.0/26, Usable range: 10.0.0.1 to 10.0.0.62, Broadcast: 10.0.0.63
2. Network: 10.0.0.64/26, Usable range: 10.0.0.65 to 10.0.0.126, Broadcast: 10.0.0.127
3. Network: 10.0.0.128/26, Usable range: 10.0.0.129 to 10.0.0.190, Broadcast: 10.0.0.191
4. Network: 10.0.0.192/26, Usable range: 10.0.0.193 to 10.0.0.254, Broadcast: 10.0.0.255


PC-PT
PC0
PC-PT
PC1
PC-PT
PC2
PC-PT
PC3
2960-24TT
Switch0
2911
Router0


Cisco Packet Tracer
File Edit Options View Tools Extensions Window Help
Logical
Physical
Switch0
Physical Config CLI Attributes
Press RETURN to get started!
%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/2, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/3, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/3, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/4, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/4, changed state to up
Switch>
Switch>
Switch>
Switch>enable
Switch#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#vlan 10
Switch(config-vlan)#vlan 20
Switch(config-vlan)#vlan 30
Switch(config-vlan)#vlan 40
Switch(config-vlan)#interface fa0/1
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 10
Switch(config-if)#interface fa0/2
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 20
Switch(config-if)#interface fa0/3
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 30
Switch(config-if)#interface fa0/4
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 40
Switch(config-if)#interface fa0/5
Switch(config-if)#switchport mode trunk
Switch(config-if)#
Copy
Paste
Top
PC-PT PC0
PC-PT PC1
PC-PT PC2
PC-PT PC3
PC-PT PC4
Realtime
Simulation


IP Configuration
DHCP
Static
IPv4 Address
10.0.0.2
Subnet Mask
255.255.255.192
Default Gateway
10.0.0.1
DNS Server
0.0.0.0

Router0

Physical Config CLI Attributes

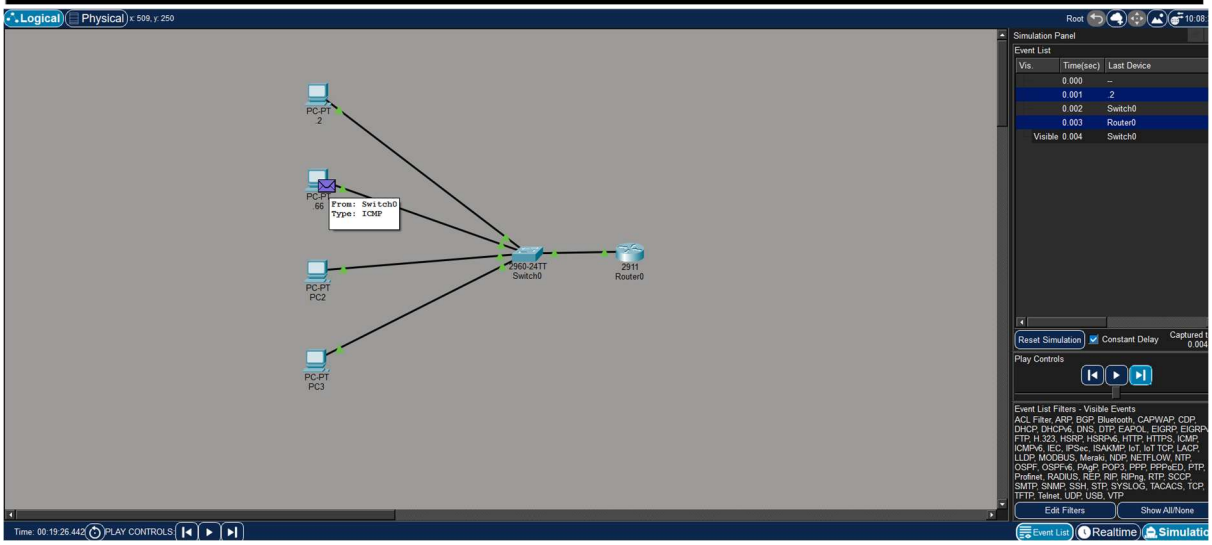
```
Router>enable
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface g0/0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

Router(config-if)#interface g0/0.10
Router(config-subif)#
%LINK-3-UPDOWN: Interface GigabitEthernet0/0.10, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.10, changed state to up
encapsulation dot1Q 10
Router(config-subif)#encapsulation dot1Q 10
Router(config-subif)#ip address 10.0.0.1 255.255.255.192
Router(config-subif)#interface g0/0.20
Router(config-subif)#
%LINK-3-UPDOWN: Interface GigabitEthernet0/0.20, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.20, changed state to up

Router(config-subif)#encapsulation dot1Q 20
Router(config-subif)#ip address 10.0.0.65 255.255.255.192
Router(config-subif)#interface g0/0.30
Router(config-subif)#
%LINK-3-UPDOWN: Interface GigabitEthernet0/0.30, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.30, changed state to up

Router(config-subif)#ip address 10.0.0.129 255.255.255.192

% Configuring IP routing on a LAN subinterface is only allowed if that
subinterface is already configured as part of an IEEE 802.10, IEEE 802.1Q,
or ISL vLAN.

Router(config-subif)#encapsulation dot1Q 20

%Configuration of multiple subinterfaces of the same main
interface with the same VID (20) is not permitted.
This VID is already configured on GigabitEthernet0/0.20.
```

Copy Paste

Top

```
Router(config-subif)#encapsulation dot1Q 30
Router(config-subif)#ip address 10.0.0.129 255.255.255.192
Router(config-subif)#interface g0/0.40
Router(config-subif)#
%LINK-3-UPDOWN: Interface GigabitEthernet0/0.40, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.40, changed state to up

Router(config-subif)#encapsulation dot1Q 40
Router(config-subif)#ip address 10.0.0.193 255.255.255.192
Router(config-subif)#
```

```
C:\>ping 10.0.0.66

Pinging 10.0.0.66 with 32 bytes of data:

Reply from 10.0.0.66: bytes=32 time=8ms TTL=127
Reply from 10.0.0.66: bytes=32 time=8ms TTL=127
Reply from 10.0.0.66: bytes=32 time=8ms TTL=127
Reply from 10.0.0.66: bytes=32 time=8ms TTL=127

Ping statistics for 10.0.0.66:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 8ms, Maximum = 8ms, Average = 8ms
```



4)

IP Address	Class	Address Range Check	Type
192.168.10.5	Class C	192.0.0.0 – 223.255.255.255	Private
172.20.15.1	Class B	128.0.0.0 – 191.255.255.255	Private
8.8.8.8	Class A	1.0.0.0 – 126.255.255.255	Public

192.168.10.5- Falls within 192.168.0.0 – 192.168.255.255 → Reserved for private networks.

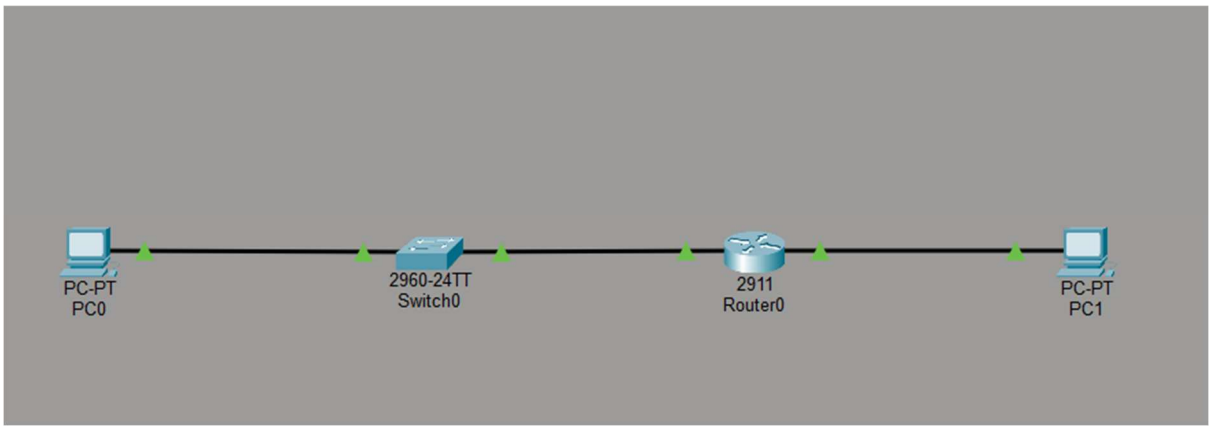
172.20.15.1- Falls within 172.16.0.0 – 172.31.255.255 → Private IP range.

8.8.8.8- Does not fall under any private IP range → Public IP (used by Google DNS).

Network Address Translation (NAT) is a technique used in networking where a router or gateway modifies IP address information in packet headers while they are in transit, allowing devices in a private network (using private IP addresses) to communicate with external networks like the internet using a single public IP address. When a device sends data, NAT replaces the private source IP with the router's public IP and maintains a mapping table to track connections; when responses return,

the router uses this table to forward packets back to the correct internal device. This process conserves public IP addresses and adds a layer of security by hiding internal network details.

5)




```
Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface gig0/0
Router(config-if)#ip address 192.168.1.1 255.255.255.0
Router(config-if)#ip nat inside
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

Router(config-if)#interface gig0/1
Router(config-if)#ip address 200.1.1.1 255.255.255.0
Router(config-if)#ip nat outside
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

Router(config-if)#access-list 1 permit 192.168.1.0 0.0.0.255
Router(config)#ip nat inside source list 1 interface gig0/1 overload
Router(config)#
Router(config)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#
```

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 200.1.1.2

Pinging 200.1.1.2 with 32 bytes of data:

Request timed out.
Reply from 200.1.1.2: bytes=32 time<1ms TTL=127
Reply from 200.1.1.2: bytes=32 time<1ms TTL=127
Reply from 200.1.1.2: bytes=32 time<1ms TTL=127

Ping statistics for 200.1.1.2:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 200.1.1.2

Pinging 200.1.1.2 with 32 bytes of data:

Reply from 200.1.1.2: bytes=32 time<1ms TTL=127
Reply from 200.1.1.2: bytes=32 time<1ms TTL=127
Reply from 200.1.1.2: bytes=32 time<1ms TTL=127
Reply from 200.1.1.2: bytes=32 time<1ms TTL=127

Ping statistics for 200.1.1.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```